

COMMENTS OF FIBER BROADBAND ASSOCIATION

March 14, 2025

The Fiber Broadband Association (“FBA”)¹ submits these comments in response to the Request for Information issued by the Networking and Information Technology Research and Development (“NITRD”) National Coordination Office (“NCO”) and National Science Foundation seeking input on the development of an Artificial Intelligence (“AI”) Action Plan.² FBA agrees that “with the right government policies, the United States can solidify its position as the leader in AI and secure a brighter future for all Americans.”³ A key component of “sustain[ing] and enhanc[ing] America’s AI dominance” will be ensuring that we have adequate infrastructure in place to support AI development and deployment.

AI requires massive data processing power, necessitating advanced data centers, the demand for which is expected to grow rapidly over the next few years.⁴ Those data centers, in turn, require connectivity to the world through enormous fiber “pipes” with “400G to 800G connectivity” – “2 to 4 times more fiber cabling than traditional hyperscale centers”⁵ – today and

¹ FBA is a not for profit trade association with more than 600 members, including telecommunications, computing, networking, system integration, engineering, and content-provider companies, as well as traditional service providers, utilities, and municipalities. Its mission is to accelerate deployment of all-fiber access networks to enable every community to leverage economic and societal benefits that only fiber can deliver. A complete list of FBA members can be found on the organization’s website: <https://www.fiberbroadband.org/>.

² National Science Foundation, *Request for Information on the Development of an Artificial Intelligence (AI) Action Plan*, 90 FR 9088-01 (Feb. 6, 2025).

³ *Id.*

⁴ See NTIA, *Request For Comments on Bolstering Data Center Growth, Resilience, and Security*, Docket No. [REDACTED] (Sept. 2024) (commenting that “data center demand is projected to grow domestically by roughly nine percent year over year through 2030”).

⁵ See “The Importance of Fiber Density in AI Data Centers,” Go!Foton (Sept. 13, 2024) available at <https://www.gofoton.com/the-importance-of-fiber-density-in-ai-data-centers/>.

even greater performance tomorrow.⁶ No other technology can match fiber’s capabilities – not only due to fiber’s superior bandwidth but also low latency and jitter that such infrastructure offers.⁷ FBA’s members are making significant investments in fiber technologies and networks that can provide this essential connectivity for many decades.⁸ However, regulatory barriers may slow the installation of fiber to data centers and increase the costs of these efforts.

For example, as part of a network build to a data center, service providers may need to cross federal lands, which could require permits from one or more federal government agencies. Permitting delays can be a significant issue for deployment, as applicants may wait months or even years to obtain all requisite approvals to access public lands and rights-of-way. The environmental review process federal agencies must undertake pursuant to the National Environmental Policy Act (“NEPA”) can similarly cause deployment delays. In the past, NEPA reviews could be quite lengthy – for instance, figures published in 2020 by the Council on Environmental Quality (“CEQ”) found that between 2010 and 2018, the average time across all federal agencies to complete an environmental impact statement was 4.5 years.⁹ FBA

⁶ Testing is already taking place for speeds up to 1.6 TB/s. *See, e.g.*, Sean Buckley, “AT&T tests 1.6 Tbps on existing Northeast long-distance fiber network,” *Lightwave* (Mar. 13, 2025), available at: <https://www.lightwaveonline.com/home/article/55274591/att-tests-16-tbps-on-existing-northeast-long-distance-fiber-network>.

⁷ *See* Comments of Fiber Broadband Association, FCC GN Docket No. 24-119 (June 6, 2024), available at: <https://www.fcc.gov/ecfs/document/10606657422018/1>. Beyond data centers, fiber-to-the-home connectivity also allows for growth in consumer AI applications such as smart home hubs, AI thermostats, security systems, smart lighting, and appliances. *See* Peter Cresse, *Accelerating AI With Fiber Systems and Strategies* (Feb. 2025), available at <https://fiberbroadband.org/resources/accelerating-ai-with-fiber-systems-and-strategies/>.

⁸ *See, e.g.*, Doug Mohny, “AI Driving Fiber-Rich Networks, Data Centers” (Nov. 29, 2023), available at: <https://fiberbroadband.org/2023/11/29/ai-driving-fiber-rich-networks-data-centers/>; Doug Mohny, “The Hunger for Data Centers in Unserved Markets” (Nov. 21, 2023), available at: <https://fiberbroadband.org/2023/11/21/the-hunger-for-data-centers-in-unserved-markets/>.

⁹ Executive Office of the President, Council on Environmental Quality, *Environmental Impact Statement Timelines (2010-2018)* (Jun. 12, 2020), available at: https://ceq.doe.gov/docs/nepa-practice/CEQ_EIS_Timeline_Report_2020-6-12.pdf.

appreciates that there have been legislative efforts to address these regulatory delays, including the 2023 amendments to NEPA to streamline environmental review processes,¹⁰ as well as introduced legislation and stated commitments to enhance efficiency in federal broadband permitting by leaders in this current Congress.¹¹ Because of the impact of federal permitting and NEPA reviews on network builds, including builds to data centers, the AI Action Plan should consider what policy changes are needed promote further efficiency in this review process to avoid unnecessary delays.

In addition to federal regulatory barriers, entities seeking to install fiber network infrastructure frequently experience permitting issues at the state and local level that can delay or derail a planned deployment. These issues include a lack of coordination and communication during the application process, excessive delays in reviewing and granting applications, and unreasonable costs and conditions imposed on applicants. Such issues could be addressed by

¹⁰ Specifically, Congress amended NEPA as part of the Fiscal Responsibility Act of 2023 by establishing a one-year timeline to complete environmental assessments, and a two-year deadline to complete environmental impact statements, subject to extensions on a case-by-case basis. PL 118-5, 137 Stat 10 (June 3, 2023). Additional reforms may be forthcoming as part of CEQ’s work to implement NEPA directives set forth in Executive Order 14154. *See* E.O. 14154, *Unleashing American Energy*, 90 FR 8353, 8355 (revoking E.O. 11991 and instructing CEQ to propose rescinding CEQ’s existing NEPA regulations and “convene a working group to coordinate the revision of agency-level implementing regulations for consistency”). CEQ has published an interim final rule to carry out these directives, and has issued guidance on NEPA implementation to federal agency and department heads. *See* Executive Office of the President, Council on Environmental Quality, *Memorandum for Heads of Federal Departments and Agencies* (Feb. 19, 2025), available at: <https://ceq.doe.gov/docs/ceq-regulations-and-guidance/CEQ-Memo-Implementation-of-NEPA-02.19.2025.pdf>.

¹¹ For example, the proposed “Federal Broadband Deployment Tracking Act” would require NTIA to submit a comprehensive plan to Congress detailing how NTIA will track the acceptance, processing, and disposal of broadband permitting applications on federal lands. *See* <https://pfluger.house.gov/news/documentsingle.aspx?DocumentID=2282#:~:text=The%20Federal%20Broadband%20Deployment%20Tracking,permitting%20applications%20on%20federal%20lands>. The need for permitting reform was also discussed at a recent hearing before the House Communications and Technology Subcommittee. *See Fixing Biden’s Broadband Blunder: Hearing Before the H. Comms. & Tech. Subcomm.*, 119th Cong. (Mar. 5, 2025) (Hearing Memo from Majority Staff).

policy reforms such as:

- establishing a single point of contact with whom applicants can interact on all necessary permitting approvals;
- ensuring clear, transparent permitting processes, including prioritization criteria, progress updates, and explicit reasons for delays or rejections;
- implementing reasonable requirements and time frames for provider submissions and government review and approval of applications;
- allocating sufficient resources to enable state and local permitting officials to process applications in a timely and cost-effective manner;
- allowing applicants to incorporate corrections to their applications along the way rather than requiring them to restart a review process anew if a permitting official finds incorrect or insufficient information at a later point in a review;
- implementing “dig once” policies to enable fiber providers to access public rights-of-way and infrastructure when costs are lower and disruption to the public is minimized;
- adopting a “deemed granted” approach for applications after a reasonable period, so long as the applicant has met reasonable requirements to submit accurate, timely, and complete information; and
- imposing requirements for access to public rights-of-way and infrastructure that are fair and reasonable and related only to and commensurate with the access provided (e.g., street restoration requirements following a fiber deployment should be reasonable and proportional to and directly arise from the deployment and should be agreed upon in advance).

The AI Action Plan should recognize the importance of such permitting reforms at the state and local level as a means of promoting efficient deployment of necessary infrastructure to support continued AI growth and development.

Respectfully Submitted,

A handwritten signature in black ink that reads "Gary Bolton". The signature is fluid and cursive, with a long horizontal stroke at the end.

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March 14, 2025

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